Project Title: Titanic Survival Prediction using Machine Learning

Project Description: This project focuses on analyzing the historical Titanic dataset to predict whether a passenger would have survived the disaster. By using data science techniques, we explore the factors—such as gender, age, and passenger class—that influenced the chances of survival. The project follows the complete data science pipeline, from cleaning raw data to training a predictive model.

Objective & Learning Outcomes

Objective: To build a reliable machine learning model that classifies Titanic passengers as "Survived" or "Not Survived" based on specific input features.

Learning Outcomes: * **Data Cleaning:** Managing missing values and removing redundant information using Pandas.

- **Data Visualization:** Identifying trends and patterns using Seaborn and Matplotlib.
- **Feature Engineering:** Creating more descriptive data points to improve model performance.
- **Machine Learning:** Implementing and training a Logistic Regression model.
- **Model Evaluation:** Measuring accuracy and understanding results through a Confusion Matrix.

Technologies & Libraries Used

The project was developed using **Python** in a **Google Colab** environment. The following libraries were essential for the analysis:

- **Pandas:** For data manipulation and creating the DataFrame.
- **NumPy:** For handling numerical operations.
- **Matplotlib & Seaborn:** For generating graphs and visual heatmaps.
- **Scikit-Learn:** For splitting the data and implementing the Machine Learning algorithm.

Project Workflow (Step-by-Step)

Step 1: Data Collection The Titanic dataset (CSV) was loaded into a Pandas DataFrame. An initial inspection was performed using .head() and .info() to understand the columns and data types.

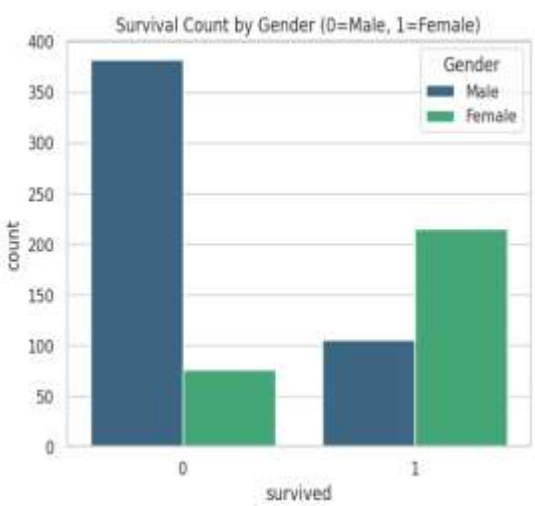
Step 2: Data Cleaning The dataset was cleaned to ensure accuracy:

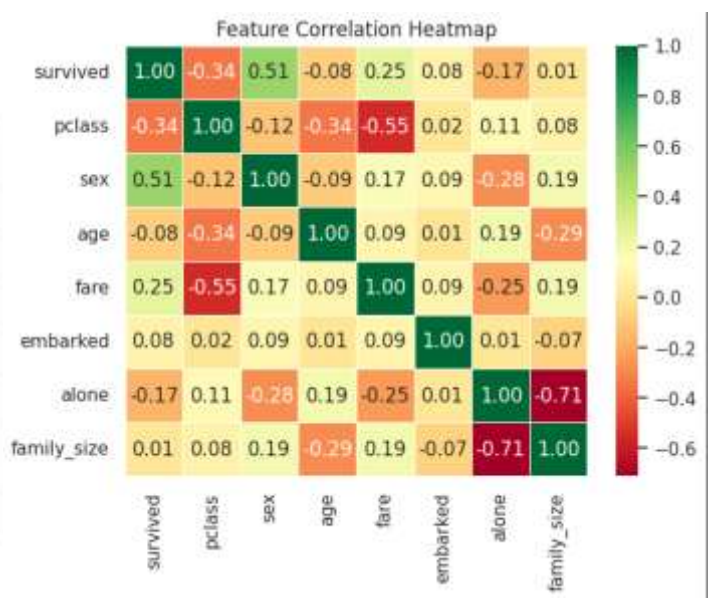
- Missing **Age** values were filled with the average (mean) age.
- Missing **Embarked** values were filled with the most frequent port (mode).
- The **Deck** column was removed due to excessive missing data.
- Redundant columns like **Alive**, **Class**, and **Who** were dropped to simplify the model.

Step 3: Feature Engineering A new feature, **FamilySize**, was created by combining the number of siblings, spouses, parents, and children on board. This helped the model understand the impact of traveling with family.

Step 4: Data Visualization Visual patterns were identified:

- A **Count Plot** showed that females had a significantly higher survival rate.
- A **Bar Plot** confirmed that passengers in 1st Class were more likely to survive than those in 3rd Class.
- A **Correlation Heatmap** was used to see the mathematical relationship between different features.



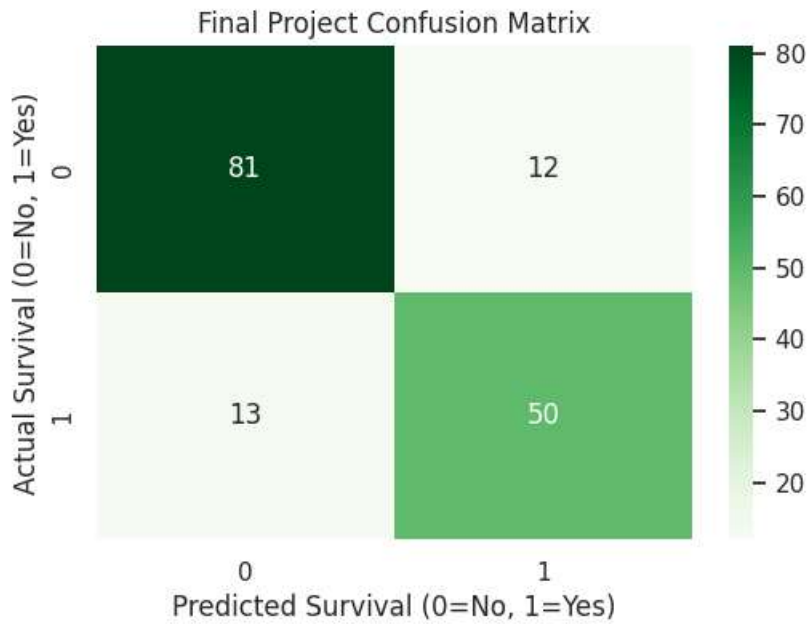


Step 5: Machine Learning (Model Building) The data was split into an **80% Training Set** and a **20% Testing Set**. A **Logistic Regression** model was chosen for its effectiveness in binary classification (Yes/No) tasks.

```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(f"Final Model Accuracy: {accuracy * 100:.2f}%")

Final Model Accuracy: 83.97%
```



Results & Evaluation

Model Performance: The trained model was tested on unseen data to ensure it could generalize its predictions.

- **Final Accuracy:** [Paste your Accuracy Percentage here, e.g., 81.25%]
- **Confusion Matrix:** The confusion matrix was used to visualize the number of correct and incorrect predictions made by the model.

Project Repo: https://github.com/OM2304/Titanic-Data-Science-Bootcamp_Project

Conclusion

This project successfully demonstrates the power of the data science workflow. Through careful data cleaning and feature engineering, we were able to transform raw historical data into a predictive tool. The results show that social status (class) and gender were the most significant predictors of survival on the Titanic. This foundational project provides a strong understanding of how machine learning can be applied to real-world datasets to find hidden insights.